

Evolution & Natural Selection

Why do living things fit their homes so well? A desert fox has big ears; an arctic fox has tiny ears. The answer is a slow process that shaped life over a very long time. This guided edition walks you through it — step by step.

 Georgia Standard S7L5

WHAT YOU'LL LEARN

- | | |
|-----------------------------------|---------------------------------|
| 01 What Is Evolution? | 05 Trait and Environment |
| 02 Variation | 06 The Math of Selection |
| 03 Natural Selection | 07 The Fossil Record |
| 04 Survival of the Fittest | 08 Review & Check |

CHAPTER WORD BANK

evolution

variation

trait

mutation

natural selection

fitness

adaptation

environment

fossil

extinction

The  boxes explain each word.

5.1 What Is Evolution?

Evolution is when the traits of a species change over many generations. It is slow. It happens to a whole group, not to one animal.

evolution = change in a species' traits over many generations.

Think about dogs. All dog breeds came from wolves. Over thousands of years, the traits changed. That is evolution.

✓ Quick Check 5.1

1. Does evolution happen to one animal, or to a whole group?
2. Is evolution fast or slow?

5.2 Variation

In any species, the members are a little bit different. These differences are called **variation**.

variation = the small differences between members of a species.

Some beetles are dark. Some are light. Some giraffes have longer necks. These small differences can decide who survives.

Why It Matters

If every animal were the same, one problem could kill them all. Variation means **some** will have a trait that helps them live.

✓ Quick Check 5.2

1. What do we call the small differences between members of a species?

5.3 Natural Selection

Natural selection is when animals with helpful traits live and have babies more than the others. Over time, the helpful trait becomes common.

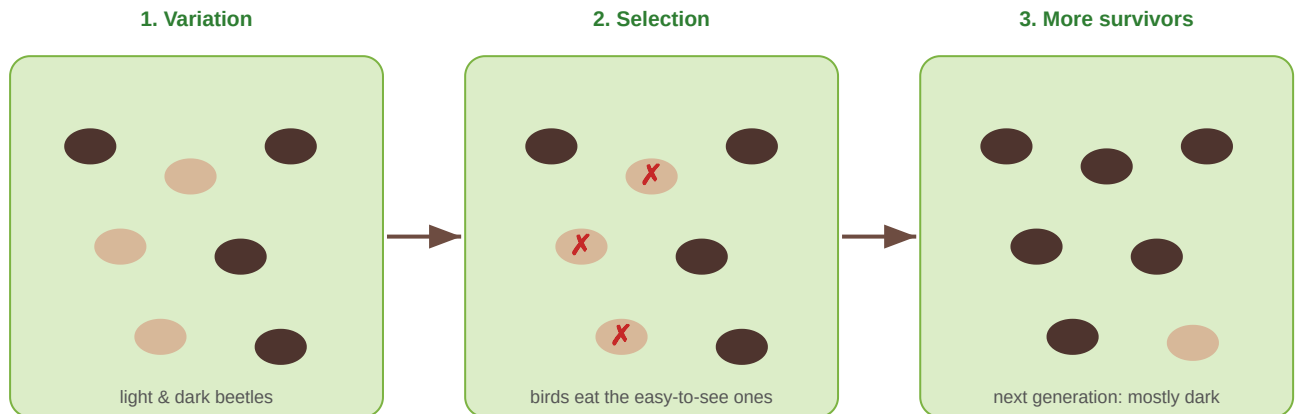


Figure 5.1 — Dark beetles hide better, survive, and become common.

★ The Beetle Story (3 steps)

- **Step 1:** Some beetles are dark, some are light (variation).
- **Step 2:** Birds eat the light ones — they are easy to see.
- **Step 3:** Dark beetles live and have babies. Soon most are dark.

Sentence Starter

The dark beetles survived because _____.

5.4 Survival of the Fittest

"Fitness" does NOT mean strongest. It means how well an animal can **survive and have babies** where it lives.

fitness = how well an animal can survive and have babies in its home.

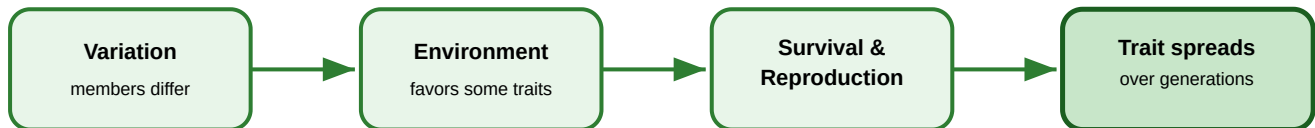
Trait	Why It Helps
Camouflage	Hides from predators
Right beak shape	Eats food more easily
Disease resistance	Survives sickness

✓ Quick Check 5.4

1. Does "fitness" mean strongest, or best at surviving and having babies?

5.5 Trait and Environment

The **environment** decides which traits help. A good trait in one place can be a bad trait in another.



Helpful traits get passed on more often — so they become more common.

Figure 5.2 — A helpful trait spreads through a group over time.

Environment	Helpful Trait
Snowy arctic	White fur
Hot desert	Big ears to cool off
Dark cave	Strong sense of smell

🔑 Same Trait, Different Result

White fur helps an arctic hare hide in snow. But white fur in a green forest would make a hare easy to see. The environment decides if a trait is good or bad.

5.6 The Math of Selection

We can use numbers to see natural selection. When a trait helps, the percent of animals with it goes **up** each generation.

100 mice live on dark rock. Dark mice hide better. Watch the dark mice grow:

Generation	Dark Mice	% Dark
1	30	30%
2	48	48%
3	66	66%
4	81	81%

★ The Pattern

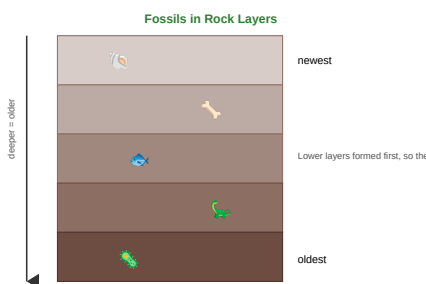
The percent of dark mice goes UP every generation. The helpful trait is spreading.

✓ Quick Check 5.6

1. As generations pass, does the percent of dark mice go up or down?

5.7 The Fossil Record

A **fossil** is the saved remains of a living thing from long ago. All the fossils together are the **fossil record** — a timeline of life.



Fossils form in rock layers. Deeper layers are **older**. The fossil record shows three things:

- **Existence:** many animals lived in the past.
- **Diversity:** life got more varied over time.
- **Extinction:** many species died out.

extinction = when a whole species dies out and is gone forever.

5.8 Review & Check

Great work — this is the last chapter! Here is everything in one place.

Big Idea	Remember This
Evolution	A species' traits change over many generations.
Variation	Small differences between members of a species.
Natural selection	Helpful traits survive and become common.
Environment	Decides which traits help. Same trait can help or hurt.
The math	A helpful trait's percent goes up each generation.
Fossil record	Shows existence, diversity, and extinction.

★ Standard Check — S7L5

You can now **(a)** use numbers to show traits changing over generations, **(b)** explain how variation and the environment lead to survival, and **(c)** describe the fossil record. 🎉 You finished the whole book!